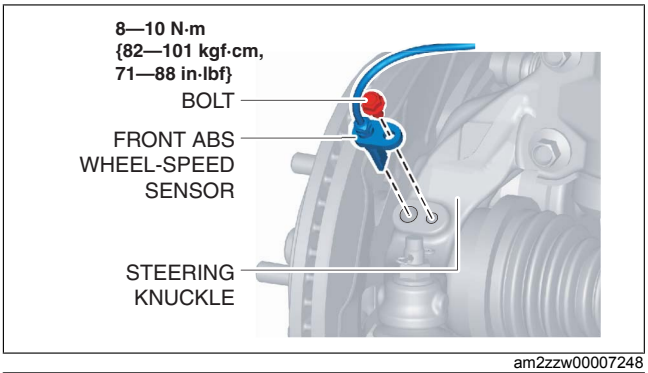
FRONT DRIVE SHAFT REMOVAL/INSTALLATION

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Caution

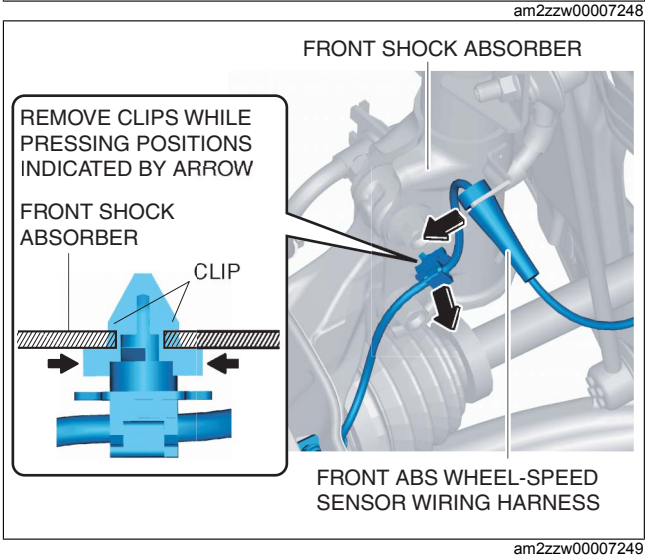
- **Performing the following procedures without first removing the ABS wheel-speed sensor may possibly cause an open circuit in the wiring harness if it is pulled by mistake. Before performing the following procedures, remove the ABS wheel-speed sensor (axle side) and fix it to an appropriate place where the sensor wiring harness will not be pulled by mistake while servicing the vehicle.**

1. Remove the front ABS wheel-speed sensor from the steering knuckle.



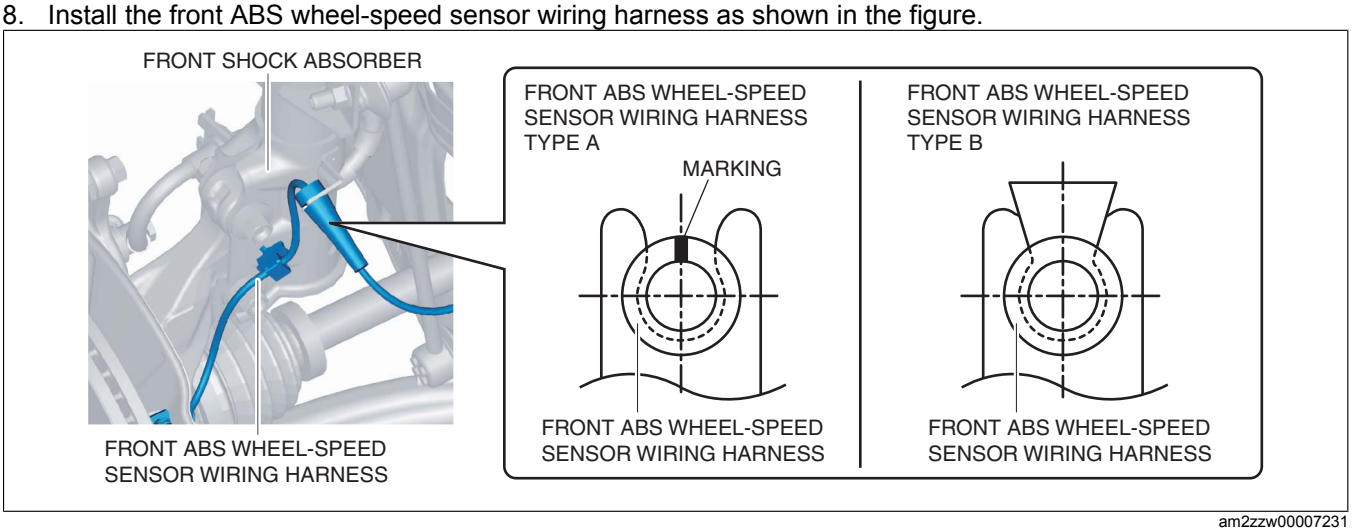
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2. Remove the front ABS wheel-speed sensor wiring harness from the front shock absorber and set it aside so that it does not interfere with the servicing.
3. Remove the front under cover No.2. (See FRONT UNDER COVER No.2 REMOVAL/INSTALLATION.)
4. Remove the splash shield. (See SPLASH SHIELD REMOVAL/INSTALLATION.)
5. Drain the manual transaxle oil or ATF. (See MANUAL TRANSAXLE OIL REPLACEMENT [F65M-R].) (See MANUAL TRANSAXLE OIL REPLACEMENT [F66M-R].) (See AUTOMATIC TRANSAXLE FLUID (ATF) REPLACEMENT [CW6A-EL].) (See AUTOMATIC TRANSAXLE FLUID (ATF) REPLACEMENT [EW6A-EL].)
6. Remove in the order indicated in the table.
7. Install in the reverse order of removal.



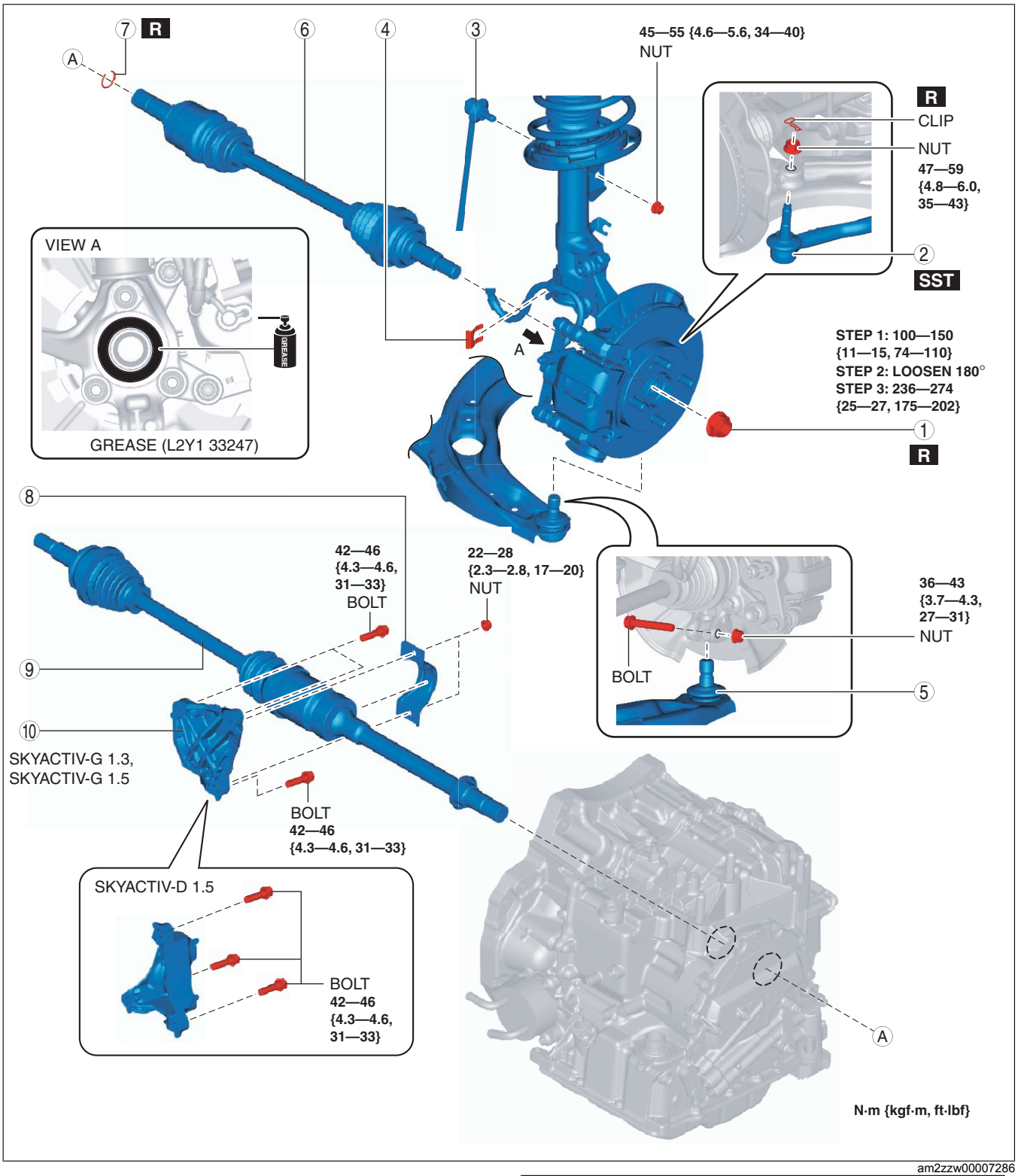
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8. Install the front ABS wheel-speed sensor wiring harness as shown in the figure.



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9. After installation, add the specified manual transaxle oil or ATF. (See MANUAL TRANSAXLE OIL REPLACEMENT [F65M-R].) (See MANUAL TRANSAXLE OIL REPLACEMENT [F66M-R].) (See AUTOMATIC TRANSAXLE FLUID (ATF) REPLACEMENT [CW6A-EL].) (See AUTOMATIC TRANSAXLE FLUID (ATF) REPLACEMENT [EW6A-EL].)



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1	Locknut (See Locknut Removal Note.) (See Locknut Installation Note.)
2	Tie-rod end (See TIE-ROD END REPLACEMENT.)
3	Stabilizer control link
4	Brake hose clip
5	Front lower arm ball joint (See FRONT LOWER ARM REMOVAL/ INSTALLATION.)

6	Front drive shaft (LH) (See Front Drive Shaft Removal Note [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See Front Drive Shaft Removal Note [SKYACTIV-D 1.5].) (See Front Drive Shaft (LH) Installation Note.)
7	Front drive shaft clip (See Front Drive Shaft Clip Installation Note.)
8	Bracket No.2 (See Bracket, Front Drive Shaft (RH) Installation Note.)

9	Front drive shaft (RH) (See Front Drive Shaft Removal Note [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See Front Drive Shaft Removal Note [SKYACTIV-D 1.5].) (See Bracket, Front Drive Shaft (RH) Installation Note.)

10	Bracket No.1 (See Bracket, Front Drive Shaft (RH) Installation Note.)

Locknut Removal Note

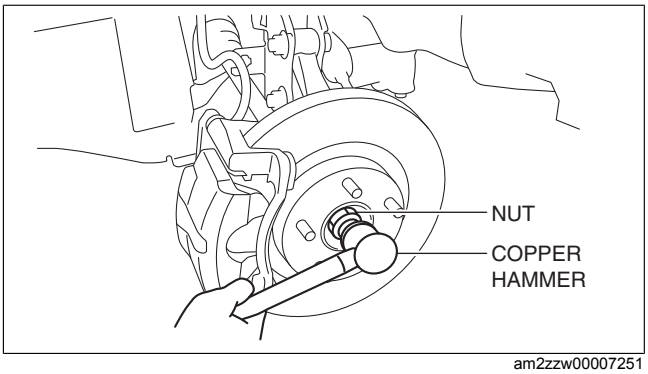
Caution
- **When removing the locknut, remove it manually without using an electric or pneumatic tool. Otherwise, the locknut may seize.**
- **When removing the locknut, do not apply load at the ground to the axle. Otherwise, it could damage the wheel hub.**

1. Remove the locknut with the brake pedal depressed.

Front Drive Shaft Removal Note [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

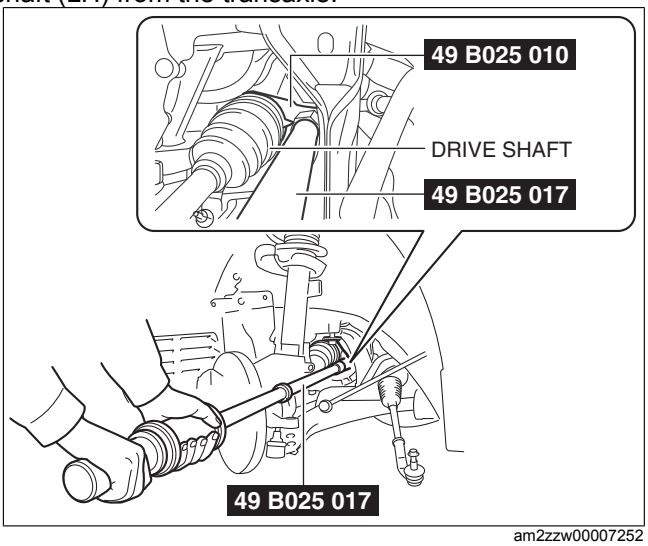
Caution
- **The oil seal at the front end of the drive shaft could be damaged. When removing the drive shaft, be careful not to damage the oil seal. If the oil seal is damaged, replace it with a new one.**

1. Install a spare nut onto the drive shaft.
2. Tap the nut with a copper hammer and separate the drive shaft from the axle.
3. Separate the drive shaft from the wheel hub.



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4. Perform the following procedures to separate the drive shaft (LH) from the transaxle.
   1. Set the **SST (49 B025 010)** to the drive shaft.
   2. Install the **SST (49 B025 017)** to the **SST (49 B025 010)**.
   3. Separate the drive shaft (LH) from the transaxle using the **SSTs**.
5. Remove the drive shaft.



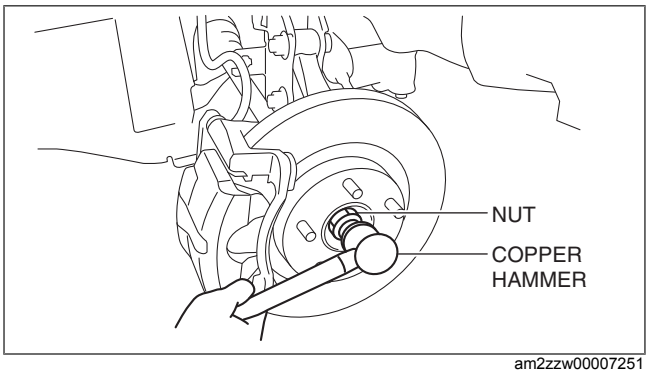
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Front Drive Shaft Removal Note [SKYACTIV-D 1.5]

Caution
- **The oil seal at the front end of the drive shaft could be damaged. When removing the drive shaft, be careful not to damage the oil seal. If the oil seal is damaged, replace it with a new one.**

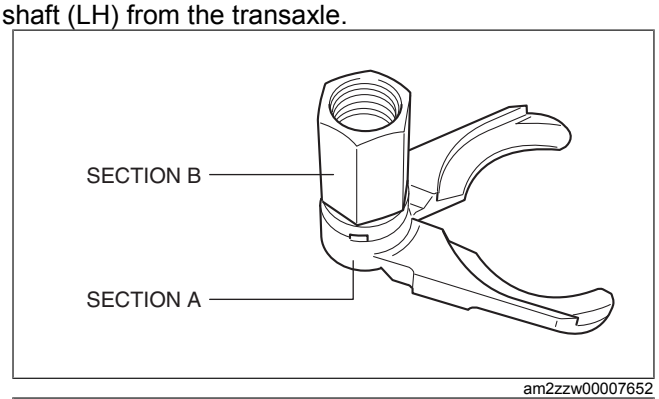
1. Install a spare nut onto the drive shaft.

2. Tap the nut with a copper hammer and separate the drive shaft from the axle.
3. Separate the drive shaft from the wheel hub.



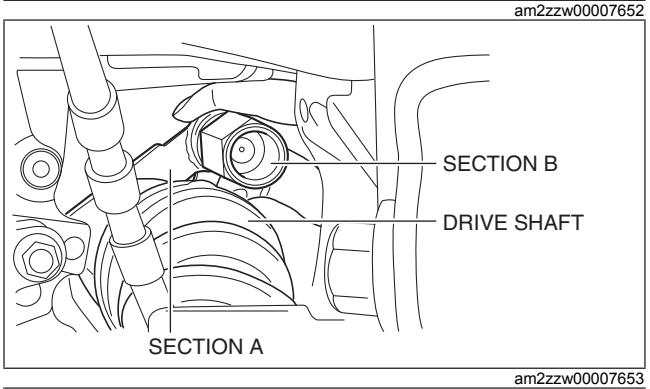
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4. Perform the following procedures to separate the drive shaft (LH) from the transaxle.
   1. Remove section B of the **SST (49 B025 010)**.



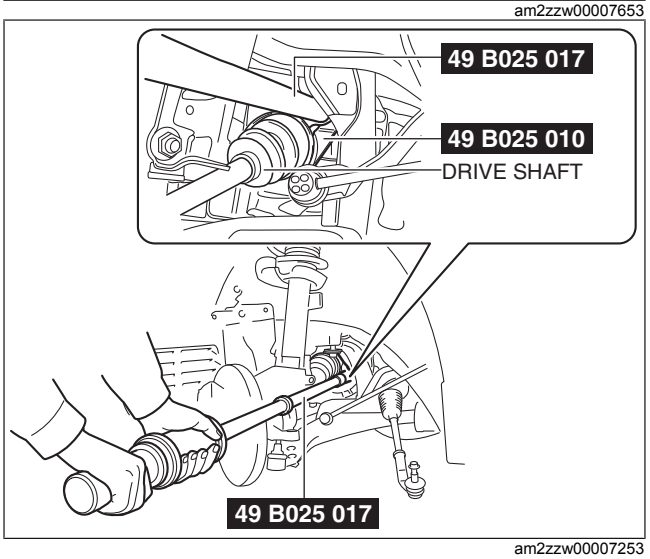
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   2. Set the drive shaft on section A of the **SST (49 B025 010)** and install section B.



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   3. Install the **SST (49 B025 017)** to the **SST (49 B025 010)**.
   4. Separate the drive shaft (LH) from the transaxle using the **SSTs**.
5. Remove the drive shaft.



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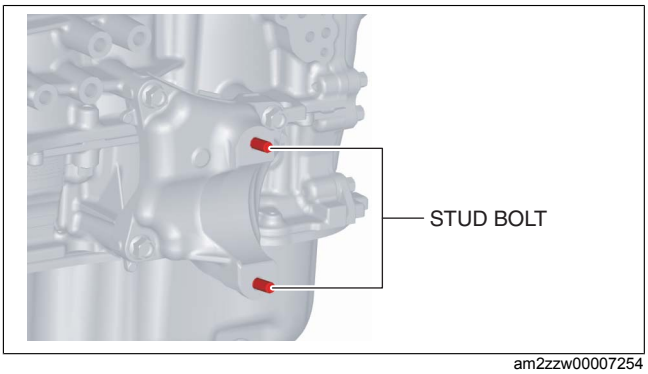
Bracket, Front Drive Shaft (RH) Installation Note

Caution
- **The oil seal at the front end of the drive shaft could be damaged. When installing the drive shaft to the transaxle, be careful not to damage the oil seal. If the oil seal is damaged, replace it with a new one.**

- **Retighten the bracket No.1 stud bolts when the bracket No.2 nuts are loosened.**

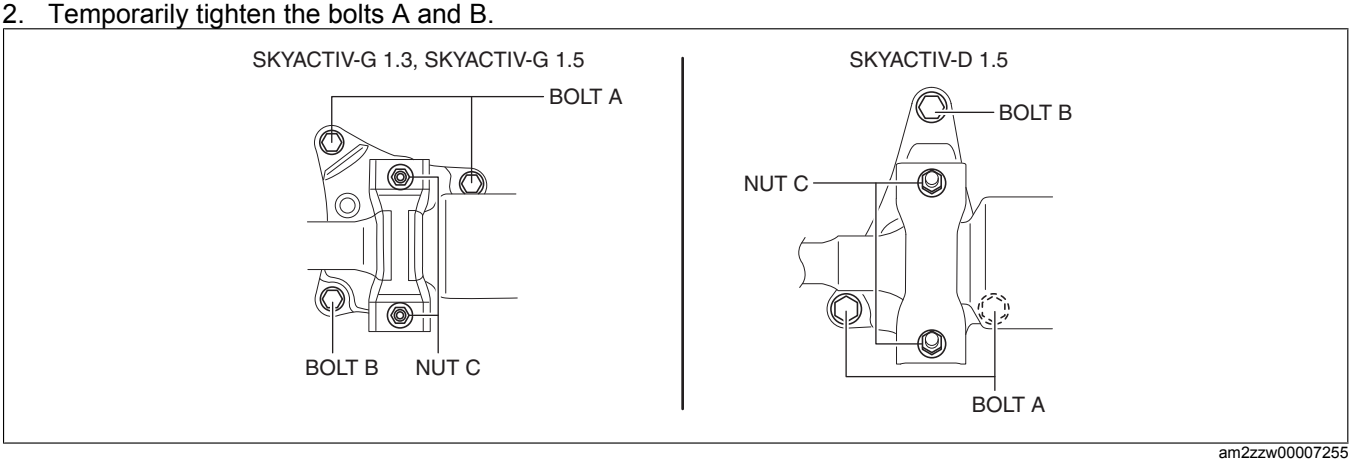
1. Tighten the bracket No.1 stud bolts.

   Tightening torque
   8—11 N·m {82—112 kgf·cm, 71—97 in·lbf}



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2. Temporarily tighten the bolts A and B.



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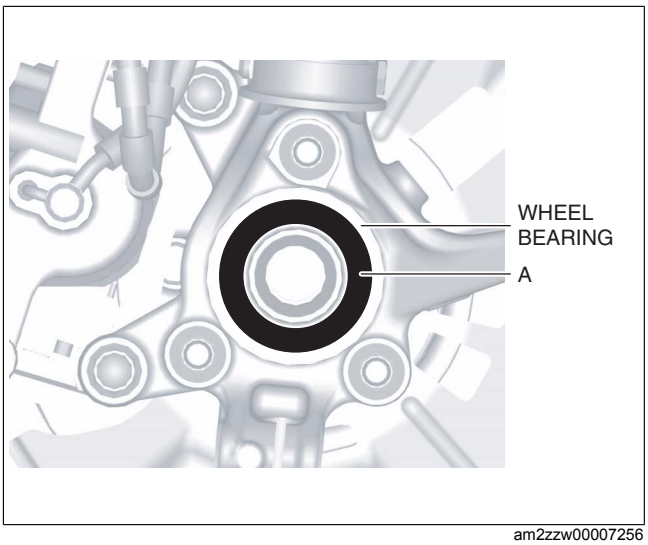
3. Tighten the bolts A, then tighten the bolt B to the specified torque.

   Tightening torque
   42—62 N·m {4.3—6.3 kgf·m, 31—45 ft·lbf}

4. Apply transaxle oil or ATF to the oil seal lip.
5. Insert the drive shaft into the transaxle until the drive shaft bearing contacts the bracket stopper.
6. Temporarily tighten the nuts C, then tighten the nuts C to the specified torque.

   Tightening torque
   22—28 N·m {2.3—2.8 kgf·m, 17—20 ft·lbf}

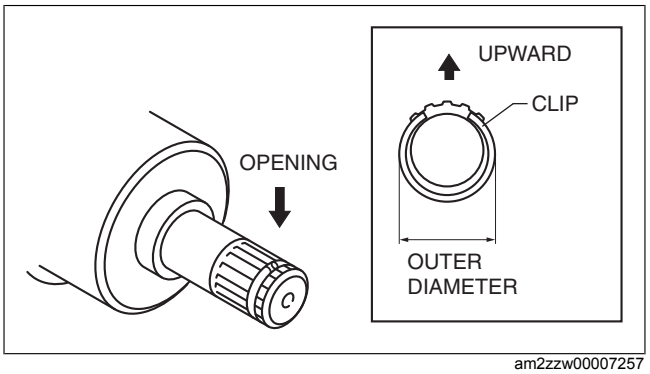
7. Apply grease (L2Y1 33247) to the wheel bearing inner race and drive shaft contact surface (Area A in figure).
8. Insert the drive shaft into the wheel hub.



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Front Drive Shaft Clip Installation Note

1. Install a new clip to the clip groove at the end of the front drive shaft with the clip opening facing upward.
2. Verify that the outer diameter of the clip is within the specification.



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Specification

Engine type	Outer diameter
SKYACTIV-G 1.3, SKYACTIV-G 1.5	Less than 28.0 mm {1.10 in}
SKYACTIV-D 1.5 (MTX)	Less than 28.0 mm {1.10 in}
SKYACTIV-D 1.5 (ATX)	Less than 34.0 mm {1.34 in}

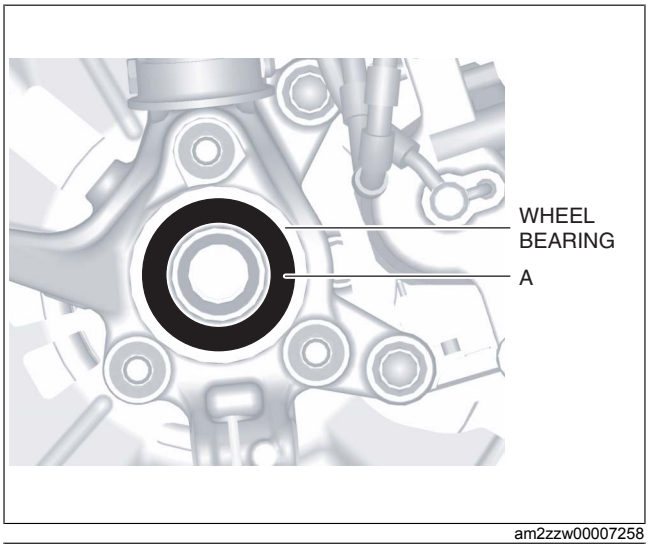
- If it exceeds the specification, repeat installation using a new clip.

Front Drive Shaft (LH) Installation Note

Caution

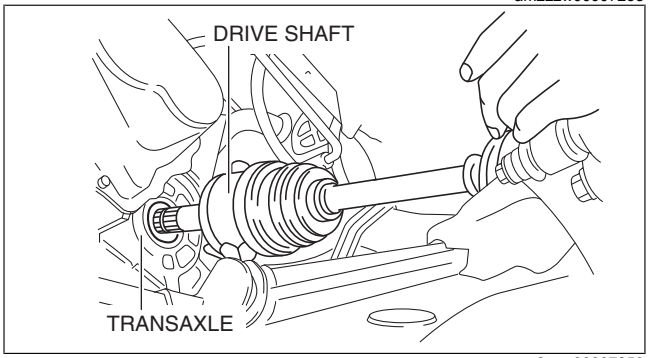
- **The oil seal at the front end of the drive shaft could be damaged. When installing the drive shaft to the transaxle, be careful not to damage the oil seal. If the oil seal is damaged, replace it with a new one.**

1. Apply transaxle oil or ATF to the oil seal lip.
2. Apply grease (L2Y1 33247) to the wheel bearing inner race and drive shaft contact surface (Area A in figure).
3. Insert the drive shaft into the wheel hub.



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4. Install the drive shaft to the transaxle.
5. After installation, pull the transaxle side outer ring forward to confirm that the drive shaft is securely held by the clip.



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Locknut Installation Note

Caution

- **When installing the locknut, install it manually without using an electric or pneumatic tool. Otherwise, the locknut may seize and the tightening force of the wheel hub may decrease and cause excessive play or abnormal noise.**
- **When installing the locknut, do not apply load at the ground to the axle. Otherwise, it could damage the wheel hub.**

1. If dust or grease is on the drive shaft thread area, wipe it off with a cloth.
2. Tighten the locknut using the following procedure and with the brake pedal depressed.
   (1) Tighten locknut to **100—150 N·m {11—15 kgf·m, 74—110 ft·lbf}** torque.
   (2) Loosen locknut to **180 degrees**.
   (3) Tighten locknut to **236—274 N·m {25—27 kgf·m, 175—202 ft·lbf}** torque.